CASE STUDY 1: PERSONAL PORTFOLIO WEBSITE

Web Development Activity-Based Learning

1. Problem Statement

Design and develop a personal portfolio webpage that presents a student's profile, education, skills, hobbies, projects, achievements, and contact details in a clear, responsive layout.

2. Objectives

• Create a structured webpage using HTML semantic sections.

• Style the page with CSS and make it usable on mobile and desktop screens.

• Provide navigation links to page sections and a simple JavaScript interaction.

• Present student information in a professional and readable format.

3. Assumptions

• The website is a static front-end demonstration and does not require a database.

• Sample details and email are placeholders and can be edited by the student.

• The user has a modern web browser; internet access is not required.

• Contact interaction displays a message only; it does not send an email.

4. Algorithm

1. Start the webpage.

2. Set the page title, viewport, and basic HTML structure.

3. Create header and navigation links for each section.

4. Add profile, education, skills, projects, hobbies, achievements, and contact sections.

5. Apply CSS colours, cards, hover transitions, and responsive rules.

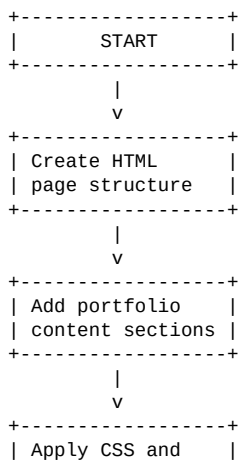
6. Use JavaScript to display a confirmation message when the contact button is clicked.

7. Open the HTML file in a browser and test the links and button.

8. End.

5. Flowchart Sketch

The following text sketch represents the main workflow:



```
| responsive rules |
+------------------+
         |
         v
+------------------+
| Add JS button    |
| interaction      |
+------------------+
         |
         v
+------------------+
| Test in browser  |
+------------------+
         |
         v
+------------------+
|        END       |
+------------------+
```

6. Pseudocode

```
BEGIN
  DISPLAY portfolio header and navigation
  DISPLAY profile and education sections
  DISPLAY technical skills and project details
  DISPLAY hobbies, achievements, and contact details

  WHEN contact button is clicked
      DISPLAY "Thank you for visiting my portfolio!"
  END WHEN
END
```

7. HTML Source Code

Save the following code using the filename shown in the heading. The file contains HTML, embedded CSS, and JavaScript.

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>My Personal Portfolio</title>
<style>
* { box-sizing: border-box; scroll-behavior: smooth; }
body { margin: 0; font-family: Arial, sans-serif; background: #f3f7fb; color: #222; line-height: 1.6; }
header { background: #14213d; color: white; text-align: center; padding: 28px 15px; }
nav { position: sticky; top: 0; background: #0077b6; padding: 12px; text-align: center; }
nav a { color: white; text-decoration: none; margin: 8px; display: inline-block; font-weight: bold; }
nav a:hover { color: #ffd166; }
section { max-width: 950px; margin: auto; padding: 28px 18px; }
h2 { color: #0077b6; }
.card { background: white; padding: 18px; margin: 12px 0; border-radius: 10px; box-shadow: 0 3px 10px
#0001; transition: transform .25s; }
.card:hover { transform: translateY(-4px); }
button { background: #0077b6; color: white; border: 0; border-radius: 5px; padding: 10px 16px; cursor:
pointer; }
footer { background: #14213d; color: white; text-align: center; padding: 15px; }
@media(max-width: 600px) { nav a { margin: 5px; font-size: 14px; } }
</style>
</head>
<body>
<header>
  <h1>My Personal Portfolio</h1>
  <p>Student | Web Developer | Technology Enthusiast</p>
</header>
<nav>
  <a href="#profile">Profile</a><a href="#education">Education</a>
  <a href="#skills">Skills</a><a href="#projects">Projects</a>
  <a href="#hobbies">Hobbies</a><a href="#contact">Contact</a>
</nav>
<section id="profile"><h2>About Me</h2><div class="card">
  <p>Hello! I am a student interested in web development, programming, and artificial intelligence. I
enjoy learning new technologies and building useful projects.</p>
</div></section>
<section id="education"><h2>Education</h2><div class="card">
  <h3>Bachelor of Technology</h3><p>Branch: Computer Science</p><p>College: SRK Institute of Technology</
p>
</div><div class="card"><h3>Intermediate</h3><p>Completed Intermediate education.</p></div></section>
<section id="skills"><h2>Technical Skills</h2><div class="card">
  <p>HTML | CSS | JavaScript | Python | Java</p>
</div></section>
<section id="projects"><h2>Projects</h2>
  <div class="card"><h3>Personal Portfolio</h3><p>A responsive website showcasing profile and skills.</
p></div>
  <div class="card"><h3>AI-Based Project</h3><p>A sample artificial intelligence project idea.</p></div>
</section>
<section id="hobbies"><h2>Hobbies and Achievements</h2><div class="card">
  <p>Hobbies: Coding, designing, learning technology, and listening to music.</p>
  <p>Achievements: Participation in technical activities and project development.</p>
</div></section>
<section id="contact"><h2>Contact</h2><div class="card">
  <p>Email: student@example.com</p><p>Location: Vijayawada, Andhra Pradesh</p>
  <button onclick="showMessage()">Contact Me</button><p id="message" aria-live="polite"></p>
</div></section>
<footer><p>&copy; 2026 My Personal Portfolio</p></footer>
<script>
function showMessage() {
  document.getElementById("message").textContent =
    "Thank you for visiting my portfolio!";
}
</script>
</body>
</html>
```

8. Test Cases

Test ID	Input / Action	Expected Result	Status
TC01	Open portfolio.html	Page loads with title and header	Pass
TC02	Click Education in navigation	Education section is brought into view	Pass
TC03	Click Contact Me	Confirmation message appears below button	Pass
TC04	Open page on a narrow/mobile screen	Navigation and content adapt to screen width	Pass

9. Sample Output

The browser displays a portfolio with a dark header, blue navigation bar, profile and education cards, skills and projects, hobbies, contact details, and a working contact-message button.

Sample browser output (illustrative layout)

> ***1: PERSONAL PORTFOLIO WEBSITE***
>
> The browser displays a portfolio with a dark header, blue navigation bar, profile and education cards, skills and projects, hobbies, contact details, and a working contact-message button.

10. Conclusion

The personal portfolio webpage was developed using HTML, CSS, and JavaScript. It organizes personal and academic information into clear sections and includes responsive styling, hover effects, navigation links, and a simple interactive button.

CASE STUDY 2: COLLEGE INFORMATION AND ADMISSIONS WEBSITE

Web Development Activity-Based Learning

1. Problem Statement

Develop a college information webpage that helps visitors find institutional information, academic departments, campus facilities, admission steps, events, and contact information.

2. Objectives

• Present college information in clearly separated sections.

• Use a navigation bar and a prominent welcome banner.

• Display departments, facilities, and events using responsive information cards.

• Provide an admission call-to-action with a basic JavaScript response.

3. Assumptions

• This is an academic prototype, not an official college website.

• Course, event, and contact information is illustrative and should be verified before publication.

• The page is static and does not process admission applications.

• The admission button displays guidance rather than submitting a form.

4. Algorithm

1. Start the webpage.

2. Create the header, navigation bar, and welcome banner.

3. Add sections for institutional information and departments.

4. Display facilities and events in responsive cards.

5. List the general admission steps and contact details.

6. Style the webpage using CSS grid, colours, spacing, and hover effects.

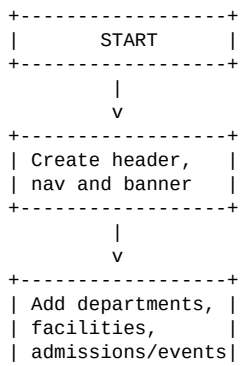
7. Add JavaScript to show a message when the admission button is clicked.

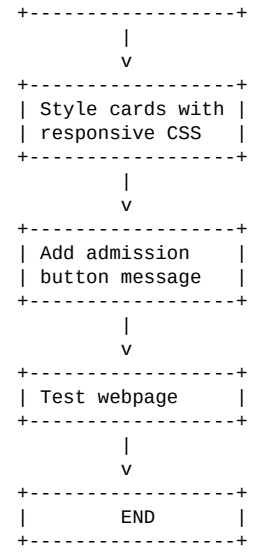
8. Test navigation and button behaviour in a browser.

9. End.

5. Flowchart Sketch

The following text sketch represents the main workflow:





6. Pseudocode

```
BEGIN
  DISPLAY college name and navigation
  DISPLAY welcome banner and admission button
  DISPLAY institutional information
  DISPLAY departments and campus facilities
  DISPLAY admission steps and college events
  DISPLAY contact details

  WHEN admission button is clicked
      DISPLAY "Please contact the college office for admission details."
  END WHEN
END
```

7. HTML Source Code

Save the following code using the filename shown in the heading. The file contains HTML, embedded CSS, and JavaScript.

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>College Information Portal</title>
<style>
* { box-sizing: border-box; }
body { margin: 0; font-family: Arial, sans-serif; background: #f4f8fc; line-height: 1.6; }
header { background: #003566; color: white; text-align: center; padding: 24px 15px; }
nav { background: #001d3d; padding: 12px; text-align: center; }
nav a { color: white; text-decoration: none; margin: 8px; display: inline-block; }
nav a:hover { color: #ffc300; }
.banner { background: linear-gradient(120deg, #0077b6, #48cae4); color: white; text-align: center;
padding: 55px 18px; }
.container { max-width: 1050px; margin: auto; padding: 28px 18px; }
h2 { color: #003566; margin-top: 0; }
.cards { display: grid; grid-template-columns: repeat(auto-fit, minmax(210px, 1fr)); gap: 16px; }
.card { background: white; padding: 18px; border-radius: 9px; box-shadow: 0 3px 10px #0001; transition:
transform .25s; }
.card:hover { transform: translateY(-4px); }
button { background: #0077b6; color: white; border: 0; border-radius: 5px; padding: 11px 16px; cursor:
pointer; }
footer { background: #001d3d; color: white; text-align: center; padding: 16px; }
@media(max-width: 600px) { .banner h2 { font-size: 25px; } }
</style>
</head>
<body>
<header><h1>SRK Institute of Technology</h1><p>Learning Today, Leading Tomorrow</p></header>
<nav>
  <a href="#about">About</a><a href="#departments">Departments</a>
  <a href="#facilities">Facilities</a><a href="#admissions">Admissions</a>
  <a href="#events">Events</a><a href="#contact">Contact</a>
</nav>
<div class="banner">
  <h2>Welcome to Our College</h2><p>Explore Knowledge, Innovation, and Excellence</p>
  <button onclick="showAdmission()">Apply for Admission</button>
  <p id="notice" aria-live="polite"></p>
</div>
<section class="container" id="about"><h2>About Our Institution</h2><div class="card">
  Our institution encourages academic learning, technical education, innovation, and student development.
</div></section>
<section class="container" id="departments"><h2>Academic Departments</h2><div class="cards">
  <div class="card"><h3>Computer Science</h3><p>Programming, software development, and computing.</p></
div>
  <div class="card"><h3>Artificial Intelligence</h3><p>Machine learning, intelligent systems, and data
science.</p></div>
  <div class="card"><h3>Electronics</h3><p>Electronic systems, circuits, and communication.</p></div>
  <div class="card"><h3>Mechanical Engineering</h3><p>Manufacturing, machines, and mechanical design.</
p></div>
</div></section>
<section class="container" id="facilities"><h2>Campus Facilities</h2><div class="cards">
  <div class="card"><h3>Library</h3><p>Learning resources and reference materials.</p></div>
  <div class="card"><h3>Computer Labs</h3><p>Practical programming and technical training.</p></div>
  <div class="card"><h3>Sports</h3><p>Facilities for games and physical activities.</p></div>
</div></section>
<section class="container" id="admissions"><h2>Admissions</h2><div class="card">
  <p>1. Review eligibility requirements.</p><p>2. Select your preferred course.</p>
  <p>3. Prepare and submit required documents.</p><p>4. Follow the official admission process.</p>
  <p><strong>Note:</strong> Confirm current requirements and deadlines with the college.</p>
</div></section>
<section class="container" id="events"><h2>College Events</h2><div class="cards">
  <div class="card"><h3>Technical Fest</h3><p>Technical competitions and innovation activities.</p></div>
  <div class="card"><h3>Workshop</h3><p>Hands-on learning and expert sessions.</p></div>
  <div class="card"><h3>Cultural Fest</h3><p>Music, dance, and student performances.</p></div>
</div></section>
<section class="container" id="contact"><h2>Contact Information</h2><div class="card">
  <p>Institution: SRK Institute of Technology</p><p>Region: Vijayawada, Andhra Pradesh</p>
  <p>Email: info@example.com (placeholder)</p>
</div></section>
<footer><p>&copy; 2026 College Information Portal</p></footer>
<script>
function showAdmission() {
  document.getElementById("notice").textContent =
    "Please contact the college office for admission details.";
```

```
}
</script>
</body>
</html>
```

8. Test Cases

Test ID	Input / Action	Expected Result	Status
TC01	Open college.html	College header and welcome banner appear	Pass
TC02	Click Facilities navigation link	Campus Facilities section is brought into view	Pass
TC03	Click Apply for Admission	Admission guidance message appears	Pass
TC04	Resize browser to mobile width	Department and facility cards reflow into fewer columns	Pass

9. Sample Output

The webpage displays the college name, navigation bar, welcome banner, academic department cards, campus facilities, admission steps, college events, and contact information. Clicking Apply for Admission displays a guidance message.

Sample browser output (illustrative layout)

> ***2: COLLEGE INFORMATION AND ADMISSIONS WEBSITE***
>
> The webpage displays the college name, navigation bar, welcome banner, academic department cards, campus facilities, admission steps, college events, and contact information. Clicking Apply for Admission displays a guidance message.

10. Conclusion

The college information and admissions webpage was developed with HTML, CSS, and JavaScript. It organizes important college details into accessible sections and uses responsive cards and an interactive admission button.

CASE STUDY 3: ONLINE LEARNING PLATFORM

Web Development Activity-Based Learning

1. Problem Statement

Build an e-learning webpage where learners can explore course categories, view course descriptions and instructor details, understand learning objectives, select an enrollment option, and view sample progress indicators.

2. Objectives

• Display course categories and popular courses using responsive cards.

• Show instructor names, descriptions, and learning objectives.

• Use JavaScript to respond to enrollment-button clicks.

• Represent sample course completion with HTML progress indicators.

3. Assumptions

• The platform is a front-end prototype with no login or backend database.

• Instructor names and progress percentages are illustrative sample data.

• Clicking Enroll Now demonstrates selection only; it does not enroll a real student.

• Progress bars are static and do not automatically update.

4. Algorithm

1. Start the webpage.

2. Create the LearnHub header and navigation.

3. Display course categories.

4. Create course cards containing titles, instructors, descriptions, and objectives.

5. Add an Enroll Now button to each course card.

6. Create progress indicators for sample courses.

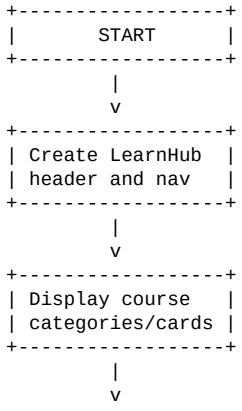
7. Use JavaScript to display the selected course when a button is clicked.

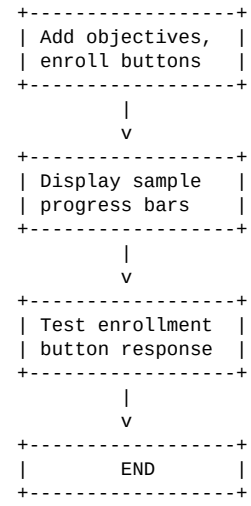
8. Test course links, buttons, and progress display in a browser.

9. End.

5. Flowchart Sketch

The following text sketch represents the main workflow:





6. Pseudocode

```
BEGIN
  DISPLAY LearnHub header and navigation
  DISPLAY course categories
  FOR each course
      DISPLAY course title, instructor, and description
      DISPLAY learning objectives and enrollment button
  END FOR
  DISPLAY sample learning progress indicators

  WHEN an enrollment button is clicked
      RECEIVE selected course name
      DISPLAY a message containing the selected course
  END WHEN
END
```

7. HTML Source Code

Save the following code using the filename shown in the heading. The file contains HTML, embedded CSS, and JavaScript.

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>LearnHub Online Learning</title>
<style>
* { box-sizing: border-box; }
body { margin: 0; font-family: Arial, sans-serif; background: #f5f7ff; color: #222; line-height: 1.5; }
header { background: #4f46e5; color: white; text-align: center; padding: 24px 15px; }
nav { background: #312e81; padding: 12px; text-align: center; }
nav a { color: white; text-decoration: none; margin: 8px; display: inline-block; }
nav a:hover { color: #facc15; }
.hero { text-align: center; padding: 38px 16px; background: #e0e7ff; }
.hero h2, h2 { color: #312e81; }
.container { max-width: 1050px; margin: auto; padding: 28px 18px; }
.courses { display: grid; grid-template-columns: repeat(auto-fit, minmax(235px, 1fr)); gap: 18px; }
.card { background: white; border-radius: 10px; overflow: hidden; box-shadow: 0 3px 10px #0001;
transition: transform .25s; }
.card:hover { transform: translateY(-4px); }
.thumbnail { background: #c7d2fe; padding: 22px; text-align: center; font-size: 38px; }
.details { padding: 17px; }
button { background: #4f46e5; color: white; border: 0; padding: 10px 14px; border-radius: 5px; cursor:
pointer; margin-top: 8px; }
button:hover { background: #312e81; }
progress { width: 100%; height: 17px; margin: 5px 0 12px; }
footer { background: #312e81; color: white; text-align: center; padding: 16px; }
</style>
</head>
<body>
<header><h1>LearnHub</h1><p>Your Online Learning Platform</p></header>
<nav><a href="#home">Home</a><a href="#categories">Categories</a><a href="#courses">Courses</a><a
href="#progress">My Progress</a></nav>
<section class="hero" id="home"><h2>Learn New Skills Online</h2>
<p>Explore programming, web development, and artificial intelligence.</p></section>
<section class="container" id="categories"><h2>Course Categories</h2><div class="courses">
  <div class="card"><div class="details"><h3>Web Development</h3><p>Learn HTML, CSS, and JavaScript.</
p></div></div>
  <div class="card"><div class="details"><h3>Programming</h3><p>Build skills using Python and Java.</p></
div></div>
  <div class="card"><div class="details"><h3>Artificial Intelligence</h3><p>Explore machine learning and
AI concepts.</p></div></div>
</div></section>
<section class="container" id="courses"><h2>Popular Courses</h2><div class="courses">
  <div class="card"><div class="thumbnail">■</div><div class="details">
    <h3>HTML &amp;amp; CSS Fundamentals</h3><p>Instructor: Alex Johnson</p><p>Create responsive webpages.</p>
    <b>Learning objectives:</b><ul><li>Understand HTML structure.</li><li>Apply CSS styling.</
li><li>Build responsive layouts.</li></ul>
    <button onclick="enroll('HTML & CSS Fundamentals')">Enroll Now</button>
  </div></div>
  <div class="card"><div class="thumbnail">■</div><div class="details">
    <h3>Python Programming</h3><p>Instructor: Sarah Lee</p><p>Learn programming and problem-solving.</p>
    <b>Learning objectives:</b><ul><li>Understand variables and data types.</li><li>Use loops and
conditions.</li><li>Write simple programs.</li></ul>
    <button onclick="enroll('Python Programming')">Enroll Now</button>
  </div></div>
  <div class="card"><div class="thumbnail">■</div><div class="details">
    <h3>Introduction to AI</h3><p>Instructor: David Miller</p><p>Discover artificial intelligence
concepts.</p>
    <b>Learning objectives:</b><ul><li>Understand AI fundamentals.</li><li>Explore machine learning.</
li><li>Identify real-world applications.</li></ul>
    <button onclick="enroll('Introduction to AI')">Enroll Now</button>
  </div></div>
</div></section>
<section class="container" id="progress"><h2>Learning Progress</h2><div class="card"><div
class="details">
  <p>HTML &amp;amp; CSS Fundamentals — 60%</p><progress value="60" max="100">60%</progress>
  <p>Python Programming — 35%</p><progress value="35" max="100">35%</progress>
  <p>Introduction to AI — 20%</p><progress value="20" max="100">20%</progress>
</div></div></section>
<footer><p>&amp;copy; 2026 LearnHub Online Learning Platform</p></footer>
<script>
function enroll(course) {
  alert("You selected: " + course + ". Enrollment is a demonstration only.");
}
```

```
</script>
</body>
</html>
```

8. Test Cases

Test ID	Input / Action	Expected Result	Status
TC01	Open learning.html	LearnHub header, navigation, and hero section appear	Pass
TC02	Click Courses in navigation	Popular Courses section is brought into view	Pass
TC03	Click Enroll Now on Python Programming	Alert names Python Programming	Pass
TC04	View Learning Progress section	Three progress bars show 60%, 35%, and 20%	Pass

9. Sample Output

The browser shows LearnHub with course categories, three course cards, instructor names, learning objectives, Enroll Now buttons, and progress bars at 60%, 35%, and 20%. Selecting a course displays a JavaScript alert.

Sample browser output (illustrative layout)

> ***3: ONLINE LEARNING PLATFORM***
>
> The browser shows LearnHub with course categories, three course cards, instructor names, learning objectives, Enroll Now buttons, and progress bars at 60%, 35%, and 20%. Selecting a course displays a JavaScript alert.

10. Conclusion

The online learning platform interface was created using HTML, CSS, and JavaScript. It demonstrates course presentation, responsive cards, learning objectives, enrollment-button interaction, and visual progress indicators.